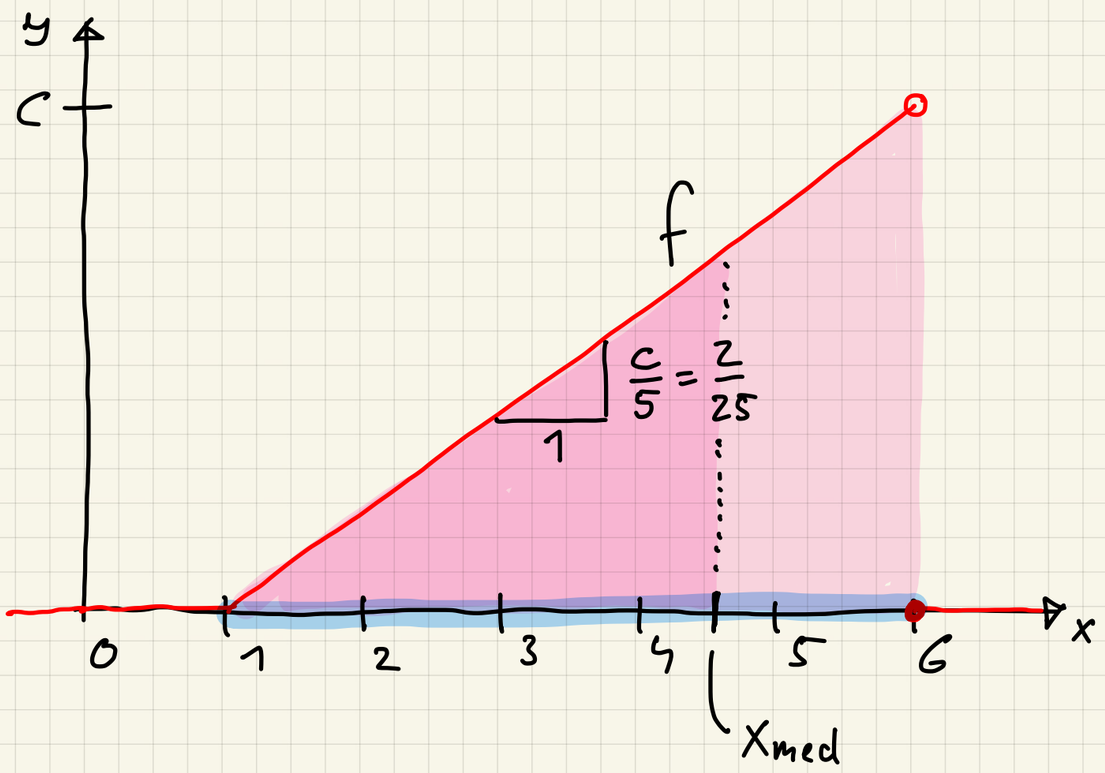


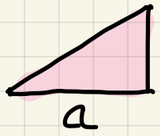
a) Funktionsgleichung

$$f(x) = \begin{cases} \frac{c}{5}(x-1) & \text{für } 1 < x < 6 \\ 0 & \text{sonst} \end{cases}$$

b) Wert von c so dass f Dichtekurve

$$\int_{-\infty}^{\infty} f(x)\,dx = \frac{1}{2} \cdot 5 \cdot c \overset{!}{=} 1 \iff c = \frac{2}{5}$$

c) x_{med} : $\int_{-\infty}^{x_{med}} f(x)\,dx = 1/2$



$\frac{1}{2} \cdot a \cdot \frac{2}{25} \cdot a = \frac{1}{25} \cdot a^2 = \frac{1}{2} \iff a = \frac{5}{2} \cdot \sqrt{2}$

$$x_{med} = 1 + \frac{5}{2}\sqrt{2} \approx 4.54$$